

WILLIAM CHANG

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ABOUT ME

Passionate Computer Science student at Columbia University with a focus on electrical engineering, computer systems, and machine learning. Programming experience in Python, Java, C, and Unity. Hands-on experience with computer vision, reward functions, and model optimization.

EDUCATION

Columbia University — B.S. Computer Science, Class of 2029

New York, NY

SKILLS

Projects

Languages: Python, Java, C, C++

Game Dev: yapples.itch.io/elemental

Hardware: Soldering, CAD, Arduino integration, circuit design

Software: Fusion360, Unity, Pandas, NumPy

AI Political Investigative Journalist: devpost.com/software/sentinel-r7oac6

AI/ML: Pytorch, TensorFlow, Computer Vision, CNNs

WORK EXPERIENCE

Electrical/Software Developer — AQUAS (CU Robotics)

2025 – Present

Develop algae-bloom clearing buoys operating in the New York area as part of the Aquas Initiative. Implement IoT technologies including network transfer via Arduino-controlled SIM commands for wireless data collection.

Electrical/Mechanical Designer — CU Airplane Club

2025 – Present

Design and build motor-powered components for the AIAA competition plane. Used Fusion360 to design a banner release mechanism and fuse-holder.

Game Developer — CU Game Development

2025 – Present

Built an action RPG in Unity as part of a 4-person team over a 4-month game incubator. Implemented puzzles, interactive structures, and world-building scripts.

Summer Intern — Inspirit AI

2022

Created a deployable CNN model to help astronomers recognize exoplanets from light-curve graphs. Extracted training-ready datasets from raw data using Pandas and NumPy; optimized models from 80% to 99.8% accuracy using PyTorch and TensorFlow.

Student Researcher — UCLA Elegant Mind Club

2024

Built a working prototype autonomous car using ML computer vision depth perception (MIDAS) with Arduino motor control. Collaborated with Professor Katsushi Arisaka to explore binocular disparity and 3D environment mapping using SLAM technology.

LEADERSHIP

Program Participant / Teaching Assistant — LeadingAI

2022 – 2023

Studied AI and machine learning alongside 50+ students. Returned as a TA to instruct 100+ students on classification, regression, and reward functions.

Student Teacher — Tsing Hua University AI and Leadership Camp

2025

Taught leadership tactics to 100+ students in Taiwan. Led discussions on cultural sensitivity, goal setting, and AI. Bridged communications across teams of 10 and organized a project presentation on improving cultural diversity at Tsing Hua.

EXTRACURRICULAR ACTIVITIES

CyberPatriot (2021 – 2025)

- Competed nationally against 6,000+ teams
- Ranked 6th worldwide in All Service category semifinals
- Achieved Platinum rank (top 5% worldwide)

OCAC International Teaching Program (2022)

- Taught English and American culture at schools in rural Taiwan
- Built a 2-week English course curriculum and led a class of 20 first-graders

ACBC Church — AI Program (2024)

- Facilitated AI-related games and activities including AI drawing and AI car racing
- Taught children about the applications of and responsible use of AI